FAKULTAS BIOLOGI
UNIVERSITAS GADJAH MADA

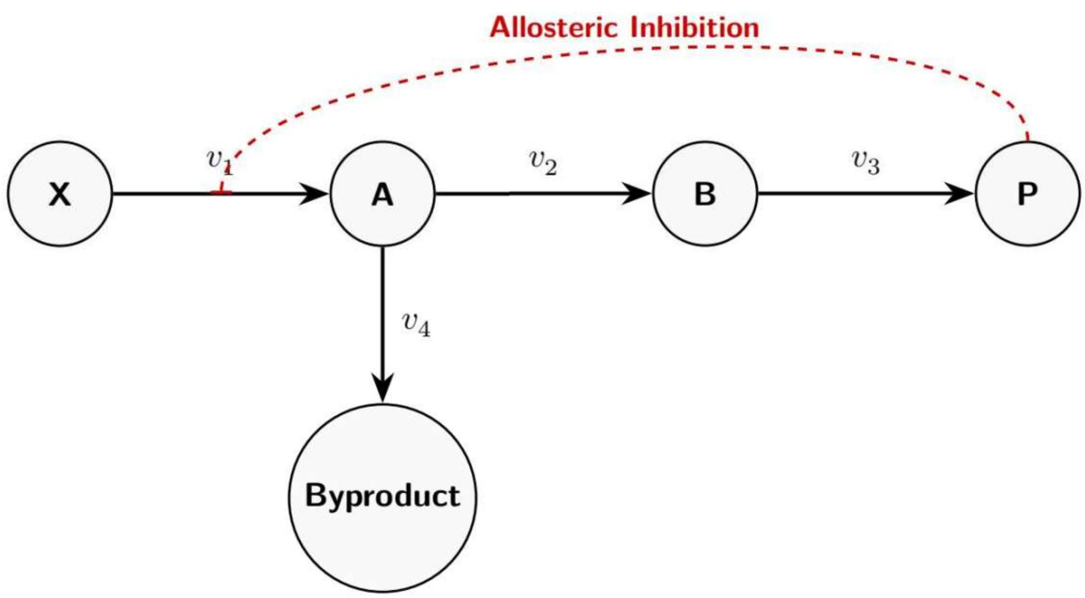
UJIAN AKHIR SEMESTER GENAP T.A. 2025/2026
MATAKULIAH BIOTEKNOLOGI (BISB211605)

Hari, Tanggal	:	-	Nama	:	
Ruang Ujian	:	-	NIM	:	
Dosen Penguji	:	Matin Nuhamunada, M.Sc., Ph.D.	No. Presensi	:	
Deadline	:	19 Juni 2026 Pukul 23.45	Tanda Tangan	:	
Sifat Ujian	:	Open Book / Take Home			

Instruksi Pengerjaan Soal:

1. **Jawablah pertanyaan berikut pada lembar jawaban yang telah disediakan (Online Form)**
2. **Anda diperbolehkan membuka buku, internet, dan sumber literasi lainnya**
3. **Anda diperbolehkan menggunakan bantuan AI (*Artificial Intelligence*) dan juga berdiskusi dengan rekan sejawat, namun jawaban akhir harus berasal dari hasil pemikiran dan *critical thinking* anda sendiri, bukan hasil dari mencontek dan plagiasi!**
4. **Anda bertanggung jawab untuk melakukan *crosscheck* dan memastikan akurasi dari hasil yang dihasilkan oleh AI**
5. **Tambahkan sitasi dalam bentuk URL jika relevan**

SYSTEMS BIOLOGY

No.	Question
	Scenario Introduction You are a metabolic engineer optimizing a microbial host to produce a high-value biochemical. Your R&D team has mapped out a critical 4-reaction pathway branch point where central metabolism is diverted toward your target product, modeled by the structural topology shown below: 

- Substrate X is maintained at a constant external concentration.
- Internal pools include intermediates A, B, and the final target Product P .
- Reaction v_4 represents an endogenous escape pathway pulling flux into an unwanted byproduct.
- The target Product P loops back to exert allosteric, non-competitive feedback inhibition on the initial commitment step (v_1).

1

QUESTION 1 - Choosing the Right Modeling Framework

As the lead engineer, you must assign computational tasks based on three distinct troubleshooting scenarios. For each scenario below, choose the most appropriate modeling approach from the options provided and provide a 1-sentence justification.

Available Frameworks:

A. Kinetic Modeling (ODE-based)
B. Constraint-Based Modeling (Flux Balance Analysis / FBA)
C. Network Topology / Graph Theory

Scenario 1: You need to calculate the maximum theoretical yield of your product across the entire genome-scale metabolic network to see if this strain is commercially viable.

Scenario 2: Your bioreactor runs are failing because intermediate B is highly toxic. You need to simulate the exact, real-time dynamic pooling of metabolite B over a 48-hour fermentation cycle to prevent cell death.

Scenario 3: You want to identify structural "hubs" or highly connected metabolite nodes in a newly sequenced host organism to ensure your gene knockouts don't accidentally disrupt essential survival pathways

ANSWER TO Q1

Scenario	Chosen Framework (A/B/C)	Justification
1		
2		
3		

2

QUESTION 2 - Quantifying the Network

Using the 4-reaction system graph (v_1, v_2, v_3, v_4) and the 3 internal metabolites (A, B, P) described in the introduction:

Q2A. Construct the Stoichiometric Matrix (S)

Fill out the stoichiometric matrix representing this system on your answer sheet. Ensure rows represent internal metabolites and columns represent flux rates.

Q2B. Formulate the Kinetic Differential Equations

Write down the baseline Ordinary Differential Equations (ODEs) representing the dynamic changes over time for all three internal pools: $\frac{d[A]}{dt}$, $\frac{d[B]}{dt}$, and $\frac{d[P]}{dt}$.

Hint: Ignore the allosteric inhibition.

Q2C. Simulate the Kinetic Model (Bonus – You can get extra points by answering this question).

In non-competitive inhibition, the inhibitor (P) binds to an allosteric site, reducing the effective $V_{1,max}$ without affecting substrate binding affinity (K_{m1}). The inhibition factor (1 + $[P]K_i$) multiplies the denominator, reducing the overall reaction velocity as $[P]$ increases. When $[P]$ = 0, the expression reduces to standard Michaelis-Menten kinetics.

$$v_1 = \frac{V_{1,max} \cdot [X]}{(K_{m1} + [X]) \cdot \left(1 + \frac{[P]}{K_i}\right)}$$

Combine the inhibition model with the baseline ODEs to simulate the system dynamics using these parameters:

Parameter	Meaning	Value
$V_{1,max}$	Max rate of V_1	5.0
K_{m1}	Michaelis constant for V_1	2.0
K_i	Inhibition constant	3.0
X	External substrate concentration	10
k_2	First-order rate constant for $A \rightarrow B$	1.0
k_3	First-order rate constant for $B \rightarrow P$	0.8
k_4	First-order rate constant for $A \rightarrow$ $byproduct$	0.3

Hint: You can utilize the example in: https://github.com/lab-biotek-bio-ugm/S1_BISB211605_Biotechnology

ANSWER TO Q2A

	v_1	v_2	v_3	v_4
Metabolite A				
Metabolite B				
Product P				

ANSWER TO Q2B

$\frac{d[A]}{dt} =$

$\frac{d[B]}{dt} =$

$\frac{d[P]}{dt} =$

ANSWER TO Q2C

Script (in Python or other languages)

Simulation Figure